

Government Engineering College Kishanganj

Department of Computer Science and Engineering

Computer Networks LAB-06

Experiment: Multi-Client File Query Server using `fork()`

Objective

Write a **TCP client-server program in C** in which multiple clients can simultaneously send queries to a server. The server must handle each connected client independently using the `fork()` system call.

Problem Statement

Assume that each client has a text file named `data.txt` stored locally. The file contains student records in the following format:

```
101 Amit CSE
102 Rahul ECE
103 Priya CSE
104 Neha EE
105 Rohit ME
```

The client should perform the following operations:

1. Read the contents of `data.txt`.
2. Ask the user to enter a **Student ID** to search.
3. Send the Student ID along with the required file data to the server.
4. Wait for the server's response.

The server should perform the following operations:

1. Create a TCP socket and wait for incoming client connections.

2. Accept connections from multiple clients.
3. Whenever a new client connects, create a child process using `fork()`.
4. Each child process should handle its corresponding client independently.
5. Search for the requested Student ID in the data received from that client.
6. Send the corresponding record back to the client.
7. The parent process should continue accepting new client connections.

Example

If the client enters:

```
Enter Student ID to search: 103
```

the server should respond:

```
Record Found: 103 Priya CSE
```

If the requested Student ID does not exist, the server should respond:

```
Record Not Found
```

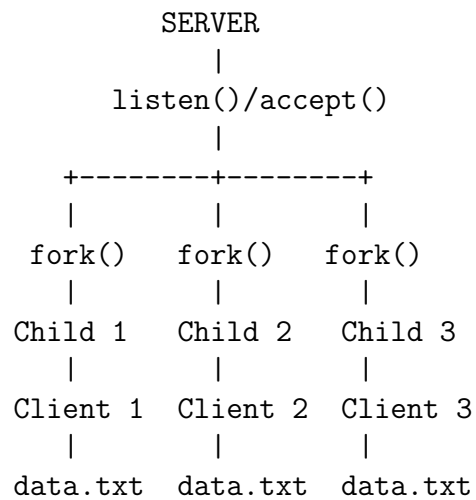
Requirements

Use the following system/library calls wherever appropriate:

```
socket(), bind(), listen(), accept(), connect(), fork(),  
read(), write(), close()
```

The parent server process must continuously accept new client connections, while each child process handles one connected client independently.

Expected Architecture



Sample Execution

Client 1:

```
Enter Student ID to search: 103
Server Response: Record Found: 103 Priya CSE
```

Client 2:

```
Enter Student ID to search: 110
Server Response: Record Not Found
```

Both clients should be able to communicate with the server **concurrently**.